A Reproducible Simulation Framework for Comparing Legislative Mechanism Bundles

Stress Tests for Productivity, Revision Moderation, Public Support, and Risk

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Legislatures need enough productivity to act, but productivity alone can also mean volatility, agenda capture, or weakly supported lawmaking. This paper presents a reproducible computational framework for comparing selected legislative mechanism bundles under shared synthetic assumptions. The simulator routes the same generated legislatures and bill streams through legislative-style processes that vary agenda access, proposal revision, voting thresholds, alternative selection, public review, lobbying safeguards, affected-group protection, package bargaining, and reversibility.

The contribution is methodological: a reusable experimental architecture for diagnosing how legislative collective decision mechanisms shift the relationship among throughput, moderating revision, support, risk, and procedural cost under controlled synthetic worlds. The framework is designed to make tradeoffs inspectable rather than to rank real institutions. It reports a diagnostic dashboard separating productivity, revision moderation, public support, low-public-support enactment, risk control, administrative feasibility, and generated public benefit per submitted proposal. A main comparison campaign demonstrates how these metrics separate across representative mechanism families, while scripted diagnostics generate seed, ablation, manipulation-stress, chamber-structure, and empirical boundary checks. The reported results are synthetic, generator-dependent design hypotheses.

CCS Concepts: • **Human-centered computing → Collaborative and social computing theory, concepts and paradigms**; • **Computing methodologies → Modeling and simulation**; • **Applied computing** → *Law*.

Additional Key Words and Phrases: legislative institutions, institutional design, computational simulation, agenda control, revision moderation, collective decision making, voting systems

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1 Introduction

The motivating question is whether systematic comparison can distinguish legislative structures that merely pass bills from those that combine productivity, revision, and representative responsiveness. A legislature can be unified without being democratic; authoritarian systems can appear cohesive because disagreement is suppressed. In this framework, representative responsiveness is operationalized through generated public support, generated public benefit, revision behavior, and burden diagnostics for affected groups.

The central modeling choice is to treat final passage thresholds as only one part of a larger institutional process that includes proposal access, agenda control, revision, review, and correction. Burden-shifting passage rules are included only as stress tests because they expose how proposal access, objection rights, information, and correction interact with final voting thresholds.

This paper treats legislative institutions as collective intelligence architectures: procedural rules determine who can contribute proposals, which signals are amplified, how expertise and public support enter decisions, how disagreement

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is resolved, and how errors are corrected. The simulator is therefore not a forecast of Congress but a tool for comparing institutional aggregation mechanisms under shared synthetic conditions.

The paper contributes four linked artifacts: a modular simulator for legislative mechanism bundles; a diagnostic dashboard separating productivity, support, risk, revision, and cost; a reproducible comparison campaign over selected mechanism families; and stress tests showing how generator assumptions, vote-kernel assumptions, and bounded manipulation probes affect the displayed tradeoffs.

The paper asks three questions. First, can a modular simulator distinguish throughput-producing rules from rules that produce moderated, broadly supported revisions under shared synthetic assumptions? Second, which mechanism families remain relevant when productivity, revision moderation, public support, risk, and administrative burden are evaluated separately? Third, which findings survive sensitivity checks on assumptions, mechanism ablations, and manipulation stressors?

This paper should be read as a framework and stress-test study, not as a validated model of Congress or a recommendation for constitutional reform. The reported results compare institutional mechanisms under shared synthetic assumptions. Empirical held-out and flow smoke tests screen for gross implausibility, but bill-topic opinion, affected-group evidence, bill-specific finance/lobbying influence, implementation outcomes, direct court/statute review, emergency court review, full statutory lineage, and cross-national institutional validation remain future work.

2 Related Work

The model draws on individual-level simulation, spatial voting, and institutional theory. Agent-based and computational political models are useful when heterogeneous actors interact under rules that create aggregate outcomes [de Marchi and Page 2014; Epstein 2006; Gilbert and Troitzsch 2005]. Spatial voting motivates one-dimensional ideal points and status-quo-relative utility [Downs 1957]. Legislative bargaining and agenda-setter models motivate explicit proposer identity and proposer gain [Baron and Ferejohn 1989; Romer and Rosenthal 1978]. Pivotal-politics and veto-player theories motivate comparing simple majority, supermajority, bicameral, and executive-veto systems [Krehbiel 1998; Tsebelis 2002]. Committee and agenda-control theory motivates treating floor denial, referral, information, and amendment control as outcomes rather than missing observations [Cox and McCubbins 2005; Gilligan and Krehbiel 1987; Krehbiel 1991; Shepsle 1979; Shepsle and Weingast 1987; Weingast and Marshall 1988].

Several tested mechanisms come from adjacent literatures. Lobbying is modeled as access, effort, information, public pressure, and participation bias rather than only direct vote buying [Austen-Smith 1993, 1995; de Figueiredo and Richter 2014; Gilens and Page 2014; Hall and Deardorff 2006; Hall and Wayman 1990]. Citizen-panel variants draw on deliberative polling and mini-public practice [Fishkin 2009; Organisation for Economic Co-operation and Development 2020]. Agenda-credit variants echo quadratic-voting and storable-vote work on expressing intensity under scarcity [Casella 2005; Lalley and Weyl 2018]. Reversibility variants draw on temporary legislation and sunset review [Gersen 2007; Ranchordas 2014]. Policy tournaments draw on Condorcet and agenda-manipulation problems [McKelvey 1976; Tideman 1987]. Model documentation follows ODD and ODD+D conventions [Grimm et al. 2010; Müller et al. 2013]. In collective intelligence terms, legislatures combine proposal generation, information filtering, preference aggregation, deliberative revision, and error correction. Existing legislative models often isolate one stage; this framework treats the stages as an interacting architecture. This paper contributes an executable comparison environment that treats institutional design as a modular collective intelligence architecture: the same synthetic actors and proposal streams are passed through different agenda, revision, voting, review, and correction mechanisms so diagnostic tradeoffs become inspectable.

3 Simulator

Each run creates a synthetic legislature, bill stream, scalar status quo, and institutional process. Legislators have ideal points $z_i \in [-1, 1]$, party labels, district preferences, and heterogeneous sensitivity to party pressure, constituency pressure, lobbying, reputation, and moderating revision. Bills have a proposer, policy position x_b, public support p_b, generated public benefit g_b, lobby pressure ℓ_b, salience, private gain, issue domain, public-benefit uncertainty, affected-group support, and concentrated harm. Enacted bills update one abstract status quo q_t, so votes evaluate policy movement rather than isolated proposals. Issue domains affect lobbying, harm, committees, and diagnostics, but the baseline model intentionally collapses policy movement into one dimension.

The voting kernel is status-quo-relative. A legislator's spatial utility is $u_i(x) = -(x - z_i)^2$. For bill b, the vote score is a weighted sum of ideology, party, public support, lobbying, revision moderation, and reputation terms:

$$s_i(b) = \sum_k w_k f_k(i, b, q_t) + \epsilon_i, \qquad \Pr(Y_i = 1) = \sigma(\max(-12, \min(12, s_i(b)))).$$

The ideology term uses $u_i(x_b) - u_i(q_t)$, party pressure uses the same comparison at the party position, constituency pressure is proportional to $2(p_b - 0.5)$, lobbying is proportional to ℓ_b, and the revision-moderation term rewards moderation, public legitimacy, and incremental policy movement. The outcome is stochastic but seed-reproducible.

Institutional modules then transform the bill lifecycle:

$$(W_t, b, q_t) \to \text{access} \to \text{revision or selection} \to \text{vote} \to \text{review} \to (q_{t+1}, m_t).$$

Modules can deny access, route to committees, force discharge or open-calendar consideration, spend attention, apply lobbying safeguards, mediate text, select among alternatives, impose harm-weighted thresholds, run citizen or objection review, apply vetoes or judicial review, and register enacted laws for later correction. Post-enactment review tracks implementation delay, capacity, underfunding, nonenforcement, and renewal pressure as separate diagnostics.

The executable loop is: generate a world; replay its bill stream through each mechanism bundle; apply the access, revision/selection, vote, review, and status-quo-update modules for each bill; then aggregate diagnostics across worlds, cases, and seeds. This paired replay is what lets the framework compare procedural changes while holding generated actors and proposals fixed.

The stylized U.S.-like conventional benchmark uses House majority passage, a Senate 60 percent threshold, presidential veto, two-thirds override, and an agenda-access gate. The gate prevents the benchmark from treating every introduced bill as a floor bill; it schedules bills using observable proxies: public support, salience, policy stability, sponsor-party fit, partisan queue conflict, capture risk, concentrated harm, and uncertainty. It does not observe generated public benefit directly.

The baseline generator is moderation-friendly by construction: generated public benefit increases as $|x_b|$ decreases, before noise, private-gain penalties, and adversarial-case modifications. That assumption makes content-selection and moderating-revision mechanisms easier to reward. The results section therefore treats the ordering as a demonstration of framework behavior under the implemented generator, and the appendix reports alternative worlds where extreme proposals can be beneficial, moderation can dilute value, lobbying can supply information, public support can be biased, and majority support can coincide with concentrated harm.

Table 1. Key generator and vote-kernel assumptions visible in the main paper. Exact formulas, bounds, scenario constants, and weights are reported in the appendix and code. These are modeling choices, not empirical estimates.

Component	Core modeling assumption	Why it matters
Legislator ideal points	One-dimensional, faction-centered distribution with heterogeneous party, constituency, lobbying, reputation, and revision-moderation sensitivities.	Determines ideological conflict and makes moderation mostly movement along one policy line.
Status quo and bills	Bills are proposer-centered policy movements evaluated against a scalar status quo; enacted bills update that status quo.	Makes voting status-quo-relative, but collapses multidimensional politics into one abstract dimension.
Public benefit	Baseline g_b is moderation-linked: $E[g_b] \approx 0.34{+}0.34(1-\lvert x_b\rvert)-0.08\, private_b$, before noise and adversarial-world changes.	This is the most important generator vulnerability because it can advantage content-selection and moderating-revision mechanisms.
Public support	Support is synthetic and correlated with generated benefit, lobbying pressure, private gain, and noise.	Public-support diagnostics are not observed public opinion and inherit generator assumptions.
Lobby/private gain	Lobby pressure is partly tied to private gain, with alternative worlds where lobbying can provide useful information.	Drives capture diagnostics and makes anti-lobbying assumptions visible.
Harm/uncertainty	Concentrated harm and uncertainty rise with distance, salience, private gain, signal mismatch, lobbying, and noise.	Drives risk screens, harm-weighted thresholds, and low-support/risk-control metrics.
Vote kernel	Legislators combine status-quo-relative spatial utility, party pressure, public support, lobbying, a revision-moderation term, reputation, and seeded noise.	The revision-moderation score is a model proxy, not a neutral moral measure of good bargaining.

Table 2. Main metrics and interpretation.

Metric	Direction	Definition	Caveat
Productivity	↑	Share of submitted bills enacted.	Can reward bad throughput.
Yield	↑	Generated public benefit per submitted proposal.	Synthetic and generator-dependent.
Revision moderation	↑	Score combining moderation, legitimacy, and incrementalism.	Not a neutral welfare measure.
Public support	↑	Enacted-bill support composite from generated support signals.	Not observed public opinion.
Low-support enactment	↓	Enacted bills below 50% generated support.	Depends on synthetic support threshold.
Risk control	↑	Inverse composite of harm, capture, low support, proposer gain, and policy shock.	Composite score.
Admin feasibility	↑	1− procedural-load index from attention, review, challenge, amendment, and selection costs.	Indirect cost proxy.

4 Metrics and Campaign

The simulator reports a dashboard rather than a single welfare score. Table 2 defines the metrics used in the main paper; secondary diagnostics remain in generated reports and appendix tables.

Yield prevents a system that blocks nearly all proposals from looking equivalent to one that produces generated public value. Risk control averages inverse low-support passage, minority harm, lobbying capture, public-preference distortion, concentrated-harm passage, proposer gain, and policy shock terms; administrative feasibility is 1− the procedural-load index.

As one sensitivity check, the paper reports a revision-moderation-weighted value profile combining revision moderation, productivity, risk control, administrative feasibility, and public support. The main interpretation relies on metric-level tradeoffs rather than this scalar score.

All main results come from one reported comparison campaign. It combines broad assumptions, adversarial generator cases, weighted party-system cases, and a stylized rising-contention timeline in one CSV. The canonical run uses 120 generated worlds per case, 101 legislators, 60 bills per run, and seed 20260428. Cases vary polarization, party loyalty,

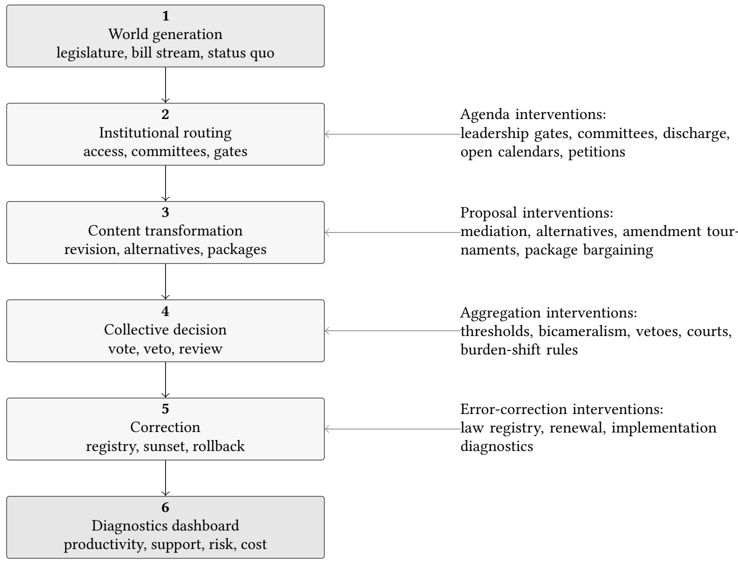


Fig. 1. Modular simulation pipeline. Each institutional bundle intervenes at one or more stages while generated legislatures, bill streams, and status-quo-relative voting remain shared across comparisons.

revision-moderation culture, lobbying, constituency pressure, party count, proposal pressure, capture pressure, and anti-lobbying reform demand. The adversarial bill-stream cases stress extremity, capture, delayed benefits, concentrated harm, and backlash dynamics.

Scenario selection follows a fixed coverage rule: include the stylized U.S.-like benchmark, simple majority, committee regular order, discharge petition, open floor calendar, a combined content-selection family, anti-capture access, harm-weighted majority, one burden-shifting stress case, one burden-shifting mediation case, and a portfolio hybrid. The appendix reports the broader scenario catalog and fixed-rule family champions.

Table 3. Mechanism families used in the main comparison. Labels match the Results figures; the appendix reports the broader implemented catalog.

Family	Labels	Mechanism theme	Primary stress risk
Conventional benchmark	CUR	House/Senate/veto gate	Bottlenecks
Simple majority	SM	Affirmative threshold	Low screening
Committee regular order	COMM	Committee-first gate	Burial/capture
Discharge petition	DIS	Committee bypass	Bypass pressure
Open floor calendar	OPEN	Default-open agenda	Floor overload
Content selection	SEL	Alternatives/tournament	Clones/agenda manipulation
Anti-capture access	ACG	Pre-floor capture screen	Gate bias
Harm-weighted majority	HARM	Affected-group threshold	Holdouts
Burden-shifting stress	DP	Pass unless blocked	False silence
Burden shift + mediation	DPM	Mediation plus challenge	Cost/weak consent
Portfolio hybrid	PORT	Combined safeguards	Complexity

The chamber-structure supplement tests representation architecture, committee power, eligibility filters, and ex ante review. It is separate because it asks how those structures shift the same productivity/revision-moderation tradeoff.

Empirical boundary checks are artifact screens, not model validation in the statistical or causal sense. `make calibrate` compares conventional baselines with broad benchmark ranges derived from Voteview, Congress.gov, govinfo, Comparative Agendas Project, lobbying-disclosure and OpenFEC records, Center for Effective Lawmaking measures, SCDB merits cases, Federal Register final-rule records, Congress.gov public-law text flags, and institutional indicators [Comparative Agendas Project 2026; Federal Election Commission 2026; Lewis et al. 2021; Library of Congress 2026; United States Government Publishing Office 2020; United States Senate Office of Public Records 2026; V-Dem Institute 2026; Volden and Wiseman 2014]. The legislative-flow layer applies one reviewed classifier to every direct action in complete pinned 116th-, 117th-, and 118th-Congress govinfo H.R. and S. BILLSTATUS archives: respectively 14,148/68,345, 15,066/72,047, and 16,213/75,239 bills/actions. A deterministic 117th-Congress bill-ID split supplies 7,564 calibration bills and 7,502 held-out bills. Only the calibration half selects the current-Congress workflow's calendar-priority threshold from a fixed 0.60–0.74 grid and 50-seed panel; floor consideration and enactment enter selection, while committee advancement is an upstream check. The selected 0.68 threshold yields committee-advance, substantive-floor, and enactment rates of 0.106, 0.061, and 0.027, compared with held-out rates of 0.101, 0.068, and 0.025; all 50 leave-one-seed-out panels reselect it. Without refitting, all three rates pass their frozen 116th-Congress backcast tolerances; the 118th forecast passes committee and floor tolerances but overpredicts enactment by 0.010825, missing its 0.010 tolerance by 0.000825. Thus 5 of 6 external cohort-metric cells pass. Classifier v3 adds successful override evidence without changing established funnel counts. Separately, a compact executive panel parses all 126,760 H.R./S. records in 22 pinned 108th–118th-Congress archives and retains 4,021 presentments. President and chamber-control fields follow the official House history, and the exact 21-veto and six-override set matches official Senate histories [Office of the Historian, U.S. House of Representatives 2026; United States Senate 2026]. The pooled empirical veto rate is 0.005223 versus 647 vetoes among 2,621 frozen simulator decisions, or 0.246852 and an exact-count ratio of 47.266. No veto tolerance was prespecified, party-control strata are noncausal, and joint resolutions are excluded, so this is a descriptive mechanism discrepancy rather than a formal held-out failure. The veto parameterization is an elevated-propensity stress mechanism rather than a U.S.-calibrated presidential-choice model.

Separate held-out and flow checks use documented empirical comparison samples for sponsors, roll calls, lobbying, campaign finance, district opinion, judicial review, implementation, statutory revision, and comparative institutions. The district-opinion path begins with a bounded Cumulative CES aggregate [Kuriwaki 2025], sponsor-district bill metadata, policy-area context, Census/ACS district context, and a 22-bill review queue. An official bill-text review retains one directionally related historical policy-preference item for H.R. 8404 and preserves 21 negative dispositions against forced policy-area matches. That retained item is joined to pinned 2012 and 2016 CCES Common Content geography and validated-voter weights [Ansolabehere and Schaffner 2012, 2016; Dagonel 2023], producing two privacy-thresholded NY-10 direct-weighted aggregates with 372 published responses. The item does not use the bill wording, both estimates predate enactment, and the layer supplies neither design-based uncertainty, MRP, affected-group evidence, causal representation, nor model validation. Across all source families, 14 targeted held-out checks pass broad prespecified ranges; the linkage audit reports all thirteen source families as linked, metadata-linked, or partially linked. Passing these screens is minimum plausibility evidence only.

Table 4. Selected empirical flow smoke tests. Raw inputs do not validate the synthetic welfare, revision moderation, harm, capture, or public-support metrics.

Signal	Raw	Simulator proxy	Does not validate
Bill attrition	enactmentRate=0.024	current-congress-workflow/productivity	public benefit or law quality
Floor consideration	floorLoad=0.067	current-congress-workflow/floor	agenda fairness or policy merit
Committee advancement from bill actions	committeeAdvanceRate=0.100	current-congress-workflow/committeeAdvanceRate	committee expertise or capture
Committee reporting from bill actions	committeeReportRate=0.064	—	committee expertise or capture
Roll-call coalition size	coalitionSize=0.604	current-system/averageEnactedSupport	public opinion
Sponsor proposal concentration	sponsorIntroductionGini=0.389	current-system/proposerAccessGini	capture or representational equity

5 Results

The selected mechanism-family comparison shows three patterns. First, throughput and support/risk diagnostics separate: mechanisms that enact more proposals can also enact more low-public-support proposals. Second, content-selection stages change the collective intelligence process by altering which alternatives enter final aggregation, not merely by changing the final yes/no threshold. Because pairwise alternatives and amendment tournaments produce nearly identical aggregate profiles in this campaign, the main text reports them as a single content-selection family. Third, portfolio safeguards reduce several risk diagnostics but increase procedural load. The stylized U.S.-like benchmark has low productivity but stronger revision moderation, risk, and support diagnostics than its directional profile suggests.

The empirical boundary checks are narrow screens for observable legislative-flow and roll-call proxies. For the richer current-Congress workflow, the full bill census now disciplines committee advancement, floor consideration, and enactment simultaneously; it does not validate generated public benefit, revision moderation, harm, capture, representation, or public-support diagnostics.

The main robustness table reports numeric generator-sensitivity, vote-kernel-ablation, and seed-stability checks. Detailed Pareto checks, ablation deltas, manipulation-stress losses, and separate PAIR/AMT rows appear in the appendix. The generator-sensitivity rows and fairness controls show that the framework can expose dependence on generator assumptions, but they do not remove that dependence. These results are mechanism-behavior hypotheses under the implemented generator, not general rankings of legislative institutions.

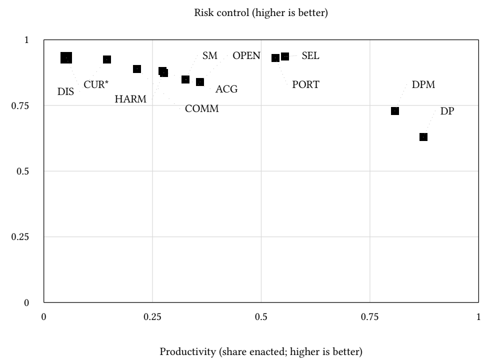
The chamber-structure supplement is reported in the appendix. It asks a separate question about representation architecture and review design, so it is not part of the ACM paper's central evidence chain.

The burden-shifting stress cases illustrate the same separation. Open burden-shifting passage has high productivity but weak revision-moderation and support diagnostics, while mediation improves some content metrics at additional procedural cost. These results are best read as failure-mode probes rather than as a central reform comparison.

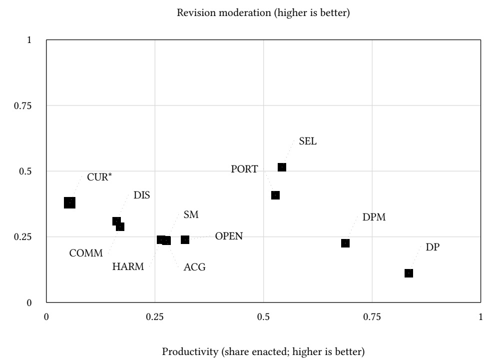
Manipulation-stress cases use bounded adversaries rather than fully co-evolving strategic parties. The adversary can choose clone or decoy proposals, noisier citizen-panel information, loose harm claims, astroturf objection pressure, proposal flooding, or anti-lobbying backlash under fixed information and budget assumptions. Stress objectives are enactment of low-support bills, blockage of high-benefit bills, or administrative overload. Appendix tables report paired losses, while adaptive worst-case search remains future work.

The portfolio variants illustrate the tradeoff between combining safeguards and increasing administrative load. They improve several risk diagnostics but do not dominate simpler alternative-selection mechanisms.

The reported ordering should therefore be read as a hypothesis about mechanism behavior under the implemented generator, not as an institutional ranking.



(a) Risk control



(b) Revision moderation

Fig. 2. Productivity tradeoff views for the representative systems. Panel (a) plots risk control across the main broad/adversarial campaign averages, with thin five-seed uncertainty bars where visible; panel (b) plots revision moderation under weighted party-system sensitivity cases. Higher is better on both y-axes. Labels are the mechanism-family labels from Table 3. The asterisk and larger square mark the stylized U.S.-like benchmark.

Table 5. Selected metric profiles for the main mechanisms.

Label	Prod.	Rev. mod.	Public	Risk	Admin feas.	Example profile
CUR*	**0.05**	**0.36**	**0.77**	**0.93**	**0.98**	**0.57**
SM	0.33	0.24	0.68	0.85	0.93	0.55
COMM	0.21	0.28	0.73	0.89	0.97	0.56
DIS	0.14	0.31	0.75	0.92	0.92	0.56
OPEN	0.36	0.25	0.67	0.84	0.90	0.55
SEL	0.55	0.50	0.79	0.94	0.71	0.67
ACG	0.27	0.24	0.70	0.88	0.94	0.55
HARM	0.28	0.26	0.69	0.87	0.93	0.55
DP	0.87	0.14	0.57	0.63	0.93	0.57
DPM	0.81	0.24	0.62	0.73	0.79	0.59
PORT	0.53	0.43	0.76	0.93	0.76	0.65

Note: Values are normalized to the metric direction shown in Table 2; grayscale shading is secondary. Admin feas. is administrative feasibility, the inverse of the cost index. Example profile is 0.30Rev. mod. +0.20Prod. +0.20Risk +0.15Admin feas. +0.15Public. The asterisk marks the stylized U.S.-like benchmark.

6 Limitations

The model is stylized. It uses one primary policy dimension, generated legislators, generated support and benefit, simplified issue domains, and abstract lobbying units. The multidimensional package scenario is a cross-issue bargaining proxy, not a full two-dimensional ideal-point model; claims about alternatives, tournaments, revision moderation, and package bargaining are therefore intentionally narrow. Public-support diagnostics distinguish salience and costly affected-group pressure from cheap amplified volume, but they are still proxies. Because public benefit is internal to the generator, welfare comparisons are conditional on model assumptions. Systems that screen extreme, uncertain, or

Table 6. Main robustness checks with numeric evidence.

Check	SEL	PORT	DP/SM	Rank ρ	Main implication
High-benefit extreme	0.69	0.72	0.57	0.95	Moderation-friendly generator is relaxed by allowing high-distance bills to carry high generated value.
Revision dilution	0.67	0.71	0.67	0.78	Moving toward the center can reduce generated public benefit.
Lobby information	0.74	0.71	0.28	0.70	Organized lobbying can supply useful technical information rather than only capture pressure.
Public-opinion error	0.68	0.68	0.41	0.93	Generated support and generated benefit are decoupled by noisy or biased support signals.
Minority-rights harm	0.66	0.64	0.19	0.57	Majority-supported bills can impose concentrated rights-like harm.
No support/revision vote	0.57	—	SM 0.49	—	Vote-kernel ablation removes public-support and revision-moderation vote inputs while retaining them as diagnostics.
Five-seed sweep	0.699–0.701	0.700–0.701	0.733–0.734	n=5	Entries show directional-score min–max across independent base seeds; ranges are small relative to mechanism-family differences.

Note: SEL and PORT entries report the example profile except in the seed row, where they report directional-score min–max. DP/SM reports default-pass low-public-support enactment for generator checks, the no-signal simple-majority profile for the vote-kernel ablation, and default-pass seed range for the seed sweep. Rank ρ is the Spearman correlation of displayed mechanism-family profile ranks against the main broad/adversarial average.

lobby-heavy bills may look better partly because the generator often encodes those features as lower expected public value. Adversarial cases partially relax this assumption, but welfare remains a synthetic signal.

The empirical layer screens conventional baselines against empirical ranges, and the held-out and flow smoke tests cover documented empirical inputs. That is enough to make several agenda-access, legislative-flow, roll-call, campaign-finance, district-opinion, review, implementation-delay, revision-text, and comparative-profile proxies inspectable, but it is not enough for model validation or parameter fitting. The linkage report currently documents complete govinfo BILLSTATUS lifecycle rows for all 14,148 116th-, 15,066 117th-, and 16,213 118th-Congress H.R. and S. bills, plus independent Congress.gov-to-govinfo agreement for 180 of 180 bounded 118th-Congress rows; bounded LDA issue context for 144 of 146 lobbying rows and 484 exact activity-text bill mentions across 26 cached public-law bill IDs; an FEC recipient-metadata join for 192 of 194 campaign-finance rows plus bounded issue, member, district, and sponsored-bill context; and sponsor-district public-law metadata for 63 of 1305 district-opinion row keys. The public-opinion review covers 66 policy-context rows across 22 bills, source-reviews official bill text for all 22, retains one historical related-issue item, preserves 21 negative dispositions, and publishes two annual privacy-thresholded aggregates for the retained item. Twenty-one bills remain proxy-only, exact or contemporaneous bill support remains absent, and affected-group support/harm rows remain at zero. Official Census TIGERweb denominators and broad ACS context cover all 22 packets across 21 sponsor districts. The report also records SCDB/Federal Register U.S.C.-section overlaps for 26 of 9341 merits rows, Federal Register document/docket metadata for 495 of 500 implementation rows, authority-text matches for 15 of 40 cached public-law rows, proposed-rule history for 23 of 51 authority-matched final-rule rows, and Congress.gov bill/action metadata for 40 of 120 public-law revision rows. The bill-law spine attaches same-policy campaign-finance sponsor context to 17 public-law rows, same-policy LDA issue/bill context to 36 public-law rows, exact official LDA filing-text bill identifiers for 26 public-law rows, exact LDA bill/action metadata context for those 26 rows, and bounded stored LDA activity-text signals; these are context, identifier, text-signal, activity-text disposition, and metadata evidence, not bill-specific influence. Separate govinfo, court-law, rulemaking-authority, rulemaking-history, bill-law, district-public-opinion policy-context, bill-item alignment, historical bill-topic support, Census denominator, ACS context, survey-source crosswalk, lobbying, campaign-finance, and bill-finance/lobbying reports expose these bounded joins and the missing evidence that remains. Exact or contemporaneous bill-topic district opinion beyond the historical related-issue pilot, bill-text-specific affected-population detail, affected-group representation, lobbying sponsor/member targeting and influence, bill-linked campaign finance, source-specific agenda and presidential-choice evidence, direct case-to-public-law review, emergency court review, fuller implementation feedback, full statutory lineage, and observed cross-national institutional productivity or fit remain validation gaps; findings that depend on those components should be read as synthetic design hypotheses. The Cumulative CES district aggregate remains a direct survey proxy. The two new annual estimates add historical related-issue district context only; they are not MRP, exact or contemporaneous bill support, affected-group harm evidence, causal representation, or model validation. The older committee and sponsor summaries are derived from sampled Congress.gov bill actions and remain rough. The three govinfo censuses replace the sample as the central legislative-flow benchmark, but strict committee reporting remains descriptive because the simulator maps only the broader ordered-reported, reported, or discharged advancement union. The OpenFEC sample supports concentration, outside-spending summaries, public recipient metadata, bounded issue-sector context, bounded candidate-member context, bounded House-candidate district context, bounded candidate-to-sponsored-bill context, public target-scope review, and cached committee/floor bill-action flags for queued rows only, not causal influence, committee-action influence, legislative-outcome causality, or bill-level capture. The LDA sample supports spending concentration, issue-taxonomy context, policy-area bill/action context, exact filing-text bill identifiers, exact bill/action metadata context, bounded stored activity-text support/position signals, a manual-review queue, a 4-row high-priority manual review slice, 102 medium-priority review packets, source-reviewed directional and position/activity packet dispositions, external current-bill mention-packet review, and cached committee/floor bill-action flags for a public-law subset only, not lobbying-contact confirmation, sponsor targeting beyond activity-text references, committee-action influence, roll-call influence, legislative-outcome causality, public benefit, or capture validation. Congress.gov public-law revision rows support only amendment, reauthorization/extension, repeal, sunset/expiration text flags, bounded bill/action metadata, same-policy finance/lobbying context, exact LDA bill identifiers and bill/action context, authority-search metadata, proposed-history/comment-portal metadata, timing metadata, bounded court U.S.C.-section overlap metadata, and the derived spine inventory, not codified lineage, complete comments, implementation outcomes, bill-specific finance/lobbying influence, or direct court invalidation. Fixture samples exercise adapter shapes but are not validation data. Strategic behavior remains bounded: proposers and lobby groups adapt, and norm erosion can raise delay, but parties, committees, coalitions, courts, agencies, media, and elections do not yet co-evolve. Chamber scenarios are sensitivity prototypes. Citizen panels, initiatives, objection windows, tournaments, audits, challenge tokens, judicial review, and law-registry review are optimistic prototypes whose costs are represented only indirectly. Package bargaining models side payments, delay, jurisdictional trades, and distributive load, not legal text or appropriations.

7 Reproducibility

The submitted manuscript is anonymized for double-blind review. The artifact is a Java 21 project with Makefile entrypoints and vendored/generated CSV summaries for the paper workflow. `make reproduce-paper-offline` is the no-network reproduction target; it regenerates the campaign, diagnostics, figures, PDFs, word-count check, anonymity scan, figure-label and table consistency checks, and PDF manifest from fixed seeds. `make supplement-anonymous` builds the double-blind supplement. Optional fixture samples can be refreshed with `make fetch-validation-samples`; they remain separate from raw empirical inputs. `make govinfo-bill-census`, `make govinfo-bill-census-116`, and `make govinfo-bill-census-118` write the three lifecycle reports; `make govinfo-executive-action-panel` verifies the compact 108th–118th-Congress panel offline; `make legislative-lifecycle-calibration` reruns the fixed calendar-capacity sweep; `make legislative-executive-action-diagnostic` compares veto pathways; `make legislative-lifecycle-temporal-replication` applies the frozen result to both external cohorts; and `make govinfo-bill-census-che` verifies all 22 executive-panel archive pins, the three lifecycle censuses, row/action integrity, stage invariants, classifier edge cases, official veto references, the preserved transport miss, and the veto discrepancy. `make empirical-linkage-report` writes the source-family linkage audit, `make empirical-linkage-roadmap` writes the corresponding upgrade gates, `make govinfo-billstatus-linkage` writes the bounded independent BILLSTATUS cross-check, `make court-law-linkage` writes the bounded SCDB/Federal Register U.S.C.-section overlap report, `make rulemaking-authority-linkage` writes the bounded Federal Register authority-search report, `make rulemaking-history-linkage` writes the bounded Federal Register proposed-history report, `make bill-law-evidence-spine` writes the bounded public-law bill/action, same-policy finance/lobbying context, exact LDA bill identifiers, and downstream timing join inventory, `make lobbying-bill-policy-context` writes the bounded LDA issue-label to cached bill/action policy-context inventory, `make lobbying-bill-mention-review` writes the official LDA filing-text bill-reference review, `make lobbying-bill-action-context` writes the exact LDA bill-identifier to public-law bill/action metadata context report, `make lobbying-bill-text-review` writes the bounded stored LDA activity-text signal review, `make lobbying-bill-disposition-review` writes the prioritized LDA disposition/target source-review queue, `make lobbying-bill-manual-disposition-review` writes the high-priority LDA manual disposition/target source review, `make lobbying-bill-medium-disposition-packets` writes the medium-priority LDA disposition/target source-review packets, `make lobbying-bill-medium-directional-packet-review` writes the medium-priority directional LDA packet review, `make lobbying-bill-medium-position-activity-packet-review` writes the medium-priority position/activity LDA packet review, `make campaign-finance-district-context` writes the bounded FEC House-candidate district-context inventory, `make campaign-finance-member-context` writes the bounded FEC candidate-to-Voteview member-context inventory, `make campaign-finance-issue-context` writes the bounded FEC transaction-label issue-context inventory, `make campaign-finance-sponsor-bill-context` writes the bounded FEC candidate-to-sponsored-bill context inventory, `make bill-finance-lobbying-campaign-finance-target-scope-review` writes the public FEC/OpenFEC target-scope review for queued campaign-finance rows, `make bill-finance-lobbying-committee-action-context` writes the cached committee/floor bill-action context for queued finance/lobbying rows, and `make validation-gap-report` writes the empirical-boundary report and appendix table. Optional raw-data rebuild targets cover the three pinned full GovInfo H.R./S. censuses, Congress.gov bill progression, the bounded GovInfo linkage cross-check, the broader Voteview/Congress.gov/LDA extract, official LDA filing-text bill-mention searches, bounded OpenFEC campaign finance, FEC campaign-finance recipient metadata, member context, issue context, and sponsored-bill context, Cumulative CES district public opinion and sponsor-district linkage, official bill-text context, pinned annual CCES historical related-issue aggregates, Census district denominators, ACS broad district context, SCDB court review and court-law U.S.C.-section overlap, Federal Register implementation, authority search, proposed-history search, and comment metadata, Congress.gov law-revision text, and Congress.gov public-law bill/action metadata when network access and any required API keys or optional data packages are available.

The minimal local smoke test is `make test`, which compiles the simulator and runs the Java test suite in under one minute on the authoring workstation. The full paper workflow typically takes several minutes because it reruns campaign, diagnostic, figure, and PDF generation. Reproduced values should match the tracked CSV/PDF manifest under the fixed seeds; raw external-data rebuilds are optional and intentionally outside the no-network path.

Use of Large Language Model Tools

The research question and core simulator idea originated with the human author. OpenAI ChatGPT/Codex tools available in the authoring environment during the April–May 2026 revision period were used for grammar and style editing, restructuring suggestions, code debugging assistance, and drafting assistance for portions of the simulator implementation, generated-report scripts, and manuscript prose. LLM tools were not used as authors. Tool outputs were edited and checked by the author; references, code outputs, experiments, tables, figures, and claims were manually reviewed against the repository artifacts before inclusion. No large language model is listed as an author.

8 Conclusion

This paper contributes a reproducible simulation framework for comparing legislative mechanism bundles as collective intelligence architectures. The reusable contribution is an experimental architecture for making legislative collective decision tradeoffs inspectable under shared synthetic worlds. The results support the framework's main methodological claim: legislative productivity, revision moderation, public support, risk control, and administrative feasibility separate under different institutional mechanisms. In the reported synthetic campaigns, content-selection stages often change the productivity/risk surface more than threshold-only baselines, while portfolio variants illustrate the cost of combining safeguards. Future work should replace synthetic proxies with stronger public-opinion, bill-linked lobbying and campaign-finance, source-specific agenda and presidential-choice, direct court-review and emergency-order, implementation, full statutory-lineage beyond bill/action metadata, observed comparative-productivity, and bargaining data.

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